RTL LABORATORY FILE

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SUBMITTED TO:

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(RTL) WEEK # 04 LAB

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RTL LAB WEEK # 04

DESIGN AND SIMULATION OF FSM USING SYSTEMVERILOG

➢ OBJECTIVE:

The objective of this lab manual is to learn and implement Register Transfer Level (RTL) designs using SystemVerilog. It aims to develop an understanding of digital systems, finite state machines (FSMs), memory modules, communication protocols, and their verification through simulation in Vivado.

LAB TASKS

➢ **Question 1:** In SystemVerilog design, synthesize, and simulate a Video Graphics Controller (VGA Controller) using SRAM.

❖ **CODE:**

- **DESIGN CODE:**

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 01
// VIDEO GRAPHICS CONTROLLER (VGA CONTROLLER) USING SRAM
module vga_controller (
    input  logic       clk,
    input  logic       rst_n,
    input  logic [7:0]  sram_data,
    output logic [16:0] sram_addr,
    output logic       sram_oe,
    output logic [3:0]  vga_r,
    output logic [3:0]  vga_g,
    output logic [3:0]  vga_b,
    output logic       vga_hsync,
    output logic       vga_vsync,
    output logic       vga_blank
);
    logic [9:0] h_count;
    logic [9:0] v_count;
    logic      h_visible;
    logic      v_visible;
    logic      video_active;
    localparam H_VISIBLE = 640;
    localparam H_FRONT   = 16;
    localparam H_SYNC    = 96;
    localparam H_BACK    = 48;
    localparam H_TOTAL   = 800;
    localparam V_VISIBLE = 480;
```

```
    localparam V_FRONT   = 10;
    localparam V_SYNC    = 2;
    localparam V_BACK    = 33;
    localparam V_TOTAL   = 525;
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            h_count <= 10'd0;
        end else if (h_count == H_TOTAL - 1) begin
            h_count <= 10'd0;
        end else begin
            h_count <= h_count + 10'd1;
        end
    end
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            v_count <= 10'd0;
        end else if (h_count == H_TOTAL - 1) begin
            if (v_count == V_TOTAL - 1) begin
                v_count <= 10'd0;
            end else begin
                v_count <= v_count + 10'd1;
            end
        end
    end
    assign h_visible = (h_count < H_VISIBLE) ? 1'b1 : 1'b0;
    assign v_visible = (v_count < V_VISIBLE) ? 1'b1 : 1'b0;
    assign video_active = h_visible && v_visible;
    assign vga_hsync = ((h_count >= (H_VISIBLE + H_FRONT)) &&
                        (h_count < (H_VISIBLE + H_FRONT + H_SYNC))) ? 1'b0 : 1'b1;
    assign vga_vsync = ((v_count >= (V_VISIBLE + V_FRONT)) &&
                        (v_count < (V_VISIBLE + V_FRONT + V_SYNC))) ? 1'b0 : 1'b1;
    assign vga_blank = ~video_active;
    assign sram_addr = video_active ? {9'd0, v_count[8:1], h_count[9:1]} : 17'd0;
    assign sram_oe = video_active;
    assign vga_r = video_active ? sram_data[7:4] : 4'd0;
    assign vga_g = video_active ? sram_data[3:2] : 4'd0;
    assign vga_b = video_active ? sram_data[1:0] : 4'd0;
endmodule
```

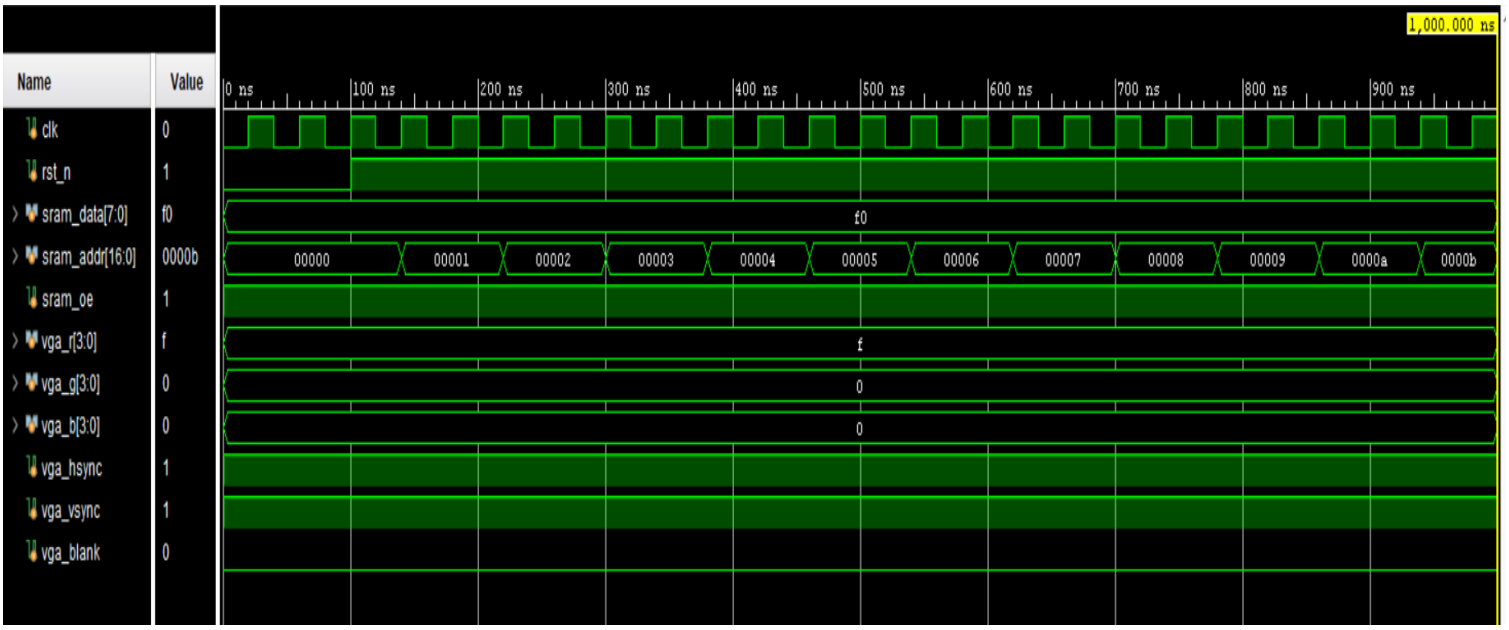
- **TEST BENCH CODE:**

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 01
// TEST-BENCH VIDEO GRAPHICS CONTROLLER (VGA CONTROLLER) USING SRAM
module tb_vga_controller;
    logic       clk;
    logic       rst_n;
    logic [7:0]  sram_data;
    logic [16:0] sram_addr;
```

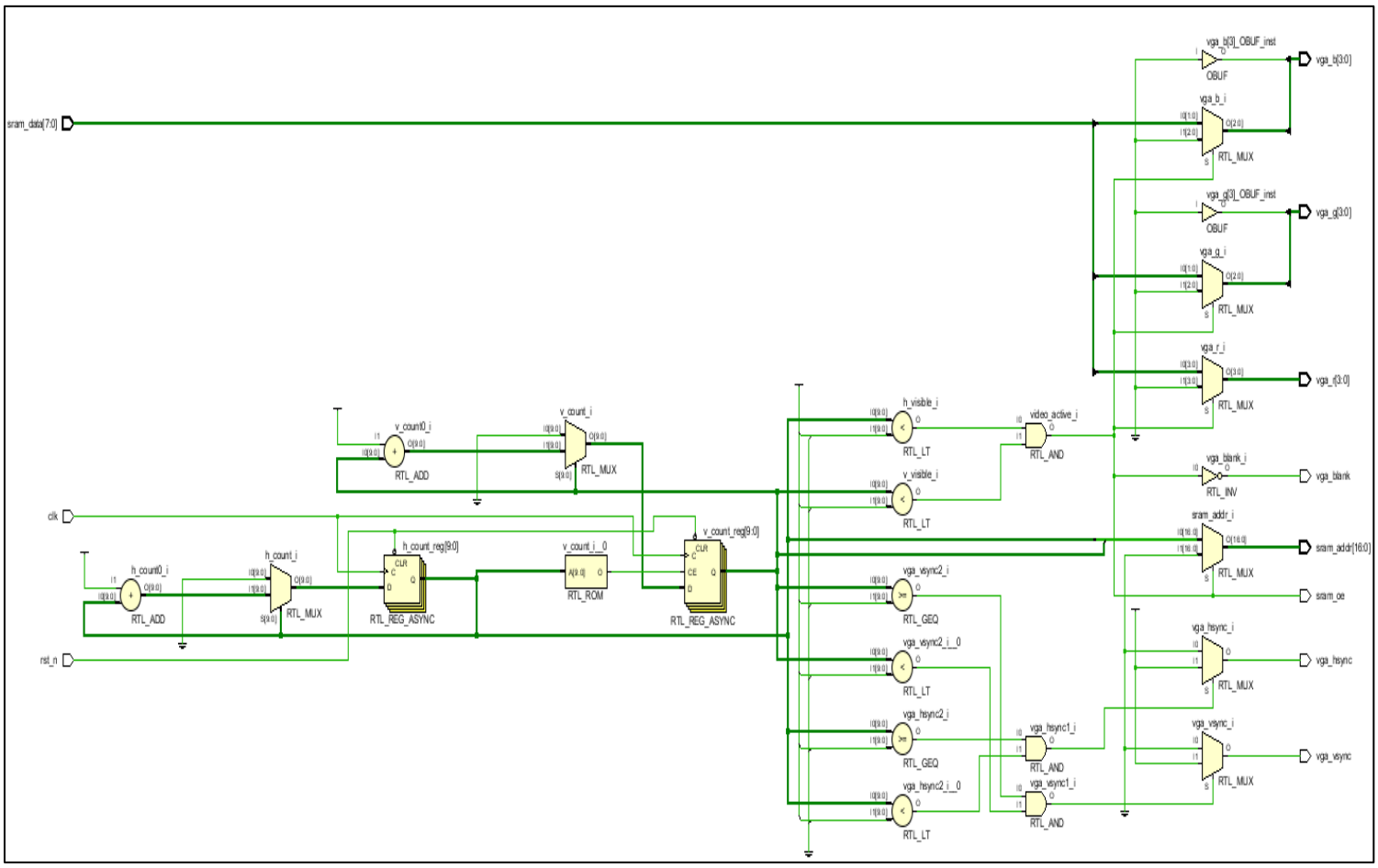
```
logic       sram_oe;
logic [3:0]  vga_r;
logic [3:0]  vga_g;
logic [3:0]  vga_b;
logic       vga_hsync;
logic       vga_vsync;
logic       vga_blank;
logic [7:0] sram_mem [0:307199];
vga_controller uut (
    .clk(clk),
    .rst_n(rst_n),
    .sram_data(sram_data),
    .sram_addr(sram_addr),
    .sram_oe(sram_oe),
    .vga_r(vga_r),
    .vga_g(vga_g),
    .vga_b(vga_b),
    .vga_hsync(vga_hsync),
    .vga_vsync(vga_vsync),
    .vga_blank(vga_blank)
);
initial begin
    clk = 0;
    forever #20 clk = ~clk;
end
initial begin
    $dumpfile("vga_controller.vcd");
    $dumpvars(0, tb_vga_controller);
end
initial begin
    integer i;
    for (i = 0; i < 307200; i++) begin
        sram_mem[i] = 8'h00;
    end
    for (i = 0; i < 6400; i++) begin
        sram_mem[i] = 8'hF0;
    end
    for (i = 6400; i < 12800; i++) begin
        sram_mem[i] = 8'h0F;
    end
end
always_comb begin
    if (sram_oe) begin
        sram_data = sram_mem[sram_addr];
    end else begin
        sram_data = 8'h00;
    end
end
initial begin
    rst_n = 1'b0;
    #100 rst_n = 1'b1;
```

```
        #2000000 $finish;
    end
endmodule
```

❖ WAVE FORM OUTPUT:



❖ SCHEMATIC OUTPUT:



- **Question 2:** Design and simulate a AXI4-Lite Master and Slave using SystemVerilog.
  - **CODE:**
  - **DESIGN CODE:**
    - **AXI_LITE_MASTER.SV:**

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 02
// AXI LITE MASTER
module axi_lite_master (
    input  logic        clk,
    input  logic        rst_n,
    input  logic        start,
    input  logic        wr_rd,
    input  logic [7:0]  addr,
    input  logic [31:0] wdata,
    output logic [31:0] rdata,
    output logic        done,
    output logic        awvalid,
    input  logic        awready,
    output logic [7:0]  awaddr,
    output logic        wvalid,
    input  logic        wready,
    output logic [31:0] wdata_out,
    output logic [3:0]  wstrb,
    output logic        bready,
    input  logic        bvalid,
    input  logic [1:0]  bresp,
    output logic        arvalid,
    input  logic        aready,
    output logic [7:0]  araddr,
    input  logic        rvalid,
    output logic        rready,
    input  logic [31:0] rdata_in,
    input  logic [1:0]  rresp
);
    typedef enum logic [1:0] {IDLE, WRITE, READ} state_t;
    state_t state, next_state;
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            state <= IDLE;
        else
            state <= next_state;
    end
    always_comb begin
        next_state = state;
        case (state)
            IDLE: begin
```

```
            if (start && wr_rd)
               next_state = WRITE;
            else if (start && !wr_rd)
               next_state = READ;
         end
         WRITE: if (bvalid && bready) next_state = IDLE;
         READ:  if (rvalid && rready) next_state = IDLE;
      endcase
   end
   always_comb begin
      awvalid = 1'b0;
      awaddr = 8'd0;
      wvalid = 1'b0;
      wdata_out = 32'd0;
      wstrb = 4'hF;
      bready = 1'b0;
      arvalid = 1'b0;
      araddr = 8'd0;
      rready = 1'b0;
      rdata = 32'd0;
      done = 1'b0;
      case (state)
         IDLE: begin
            awaddr = addr;
            araddr = addr;
            wdata_out = wdata;
         end
         WRITE: begin
            awvalid = 1'b1;
            awaddr = addr;
            if (awready) begin
               wvalid = 1'b1;
               wdata_out = wdata;
               if (wready) begin
                  bready = 1'b1;
                  if (bvalid && bready) begin
                     done = 1'b1;
                  end
               end
            end
         end
         READ: begin
            arvalid = 1'b1;
            araddr = addr;
            if (arready) begin
               rready = 1'b1;
               if (rvalid && rready) begin
                  rdata = rdata_in;
                  done = 1'b1;
               end
            end
```

```
        end
      endcase
   end
endmodule
```

➢ AXI_LITE_SLAVE.SV:

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 02
// AXI LITE SLAVE
module axi_lite_slave (
   input  logic       clk,
   input  logic       rst_n,
   input  logic       awvalid,
   output logic       awready,
   input  logic [7:0]  awaddr,
   input  logic       wvalid,
   output logic       wready,
   input  logic [31:0] wdata,
   input  logic [3:0]  wstrb,
   input  logic       bready,
   output logic       bvalid,
   output logic [1:0]  bresp,
   input  logic       arvalid,
   output logic       arready,
   input  logic [7:0]  araddr,
   output logic       rvalid,
   input  logic       rready,
   output logic [31:0] rdata,
   output logic [1:0]  rresp
);
   logic [31:0] mem [0:255];
   typedef enum logic [1:0] {IDLE, WR_RESP, RD_RESP} state_t;
   state_t state, next_state;
   always_ff @(posedge clk or negedge rst_n) begin
      if (!rst_n)
         state <= IDLE;
      else
         state <= next_state;
   end
   always_comb begin
      next_state = state;
      case (state)
         IDLE: begin
            if (awvalid && wvalid)
               next_state = WR_RESP;
            else if (arvalid)
               next_state = RD_RESP;
         end
```

```
            WR_RESP: if (bready) next_state = IDLE;
            RD_RESP: if (rready) next_state = IDLE;
        endcase
    end
    always_comb begin
        awready = 1'b0;
        wready = 1'b0;
        bvalid = 1'b0;
        bresp = 2'b00;
        arready = 1'b0;
        rvalid = 1'b0;
        rdata = 32'd0;
        rresp = 2'b00;
        case (state)
            IDLE: begin
                awready = 1'b1;
                arready = 1'b1;
            end
            WR_RESP: begin
                if (awvalid && wvalid) begin
                    mem[awaddr] = wdata;
                    bvalid = 1'b1;
                    bresp = 2'b00;
                end
            end
            RD_RESP: begin
                if (arvalid) begin
                    rdata = mem[araddr];
                    rvalid = 1'b1;
                    rresp = 2'b00;
                end
            end
        endcase
    end
endmodule
```

➢ TOP_AXI_LITE.SV:

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 02
// TOP AXI LITE
module top_axi_lite (
    input  logic        clk,
    input  logic        rst_n,
    input  logic        start,
    input  logic        wr_rd,
    input  logic [7:0]  addr,
    input  logic [31:0] wdata,
    output logic [31:0] rdata,
```

```
    output logic        done
);
    logic       awvalid;
    logic       awready;
    logic [7:0]  awaddr;
    logic       wvalid;
    logic       wready;
    logic [31:0] wdata_out;
    logic [3:0]  wstrb;
    logic       bready;
    logic       bvalid;
    logic [1:0]  bresp;
    logic       arvalid;
    logic       arready;
    logic [7:0]  araddr;
    logic       rvalid;
    logic       rready;
    logic [31:0] rdata_in;
    logic [1:0]  rresp;
    axi_lite_master master (
        .clk(clk),
        .rst_n(rst_n),
        .start(start),
        .wr_rd(wr_rd),
        .addr(addr),
        .wdata(wdata),
        .rdata(rdata),
        .done(done),
        .awvalid(awvalid),
        .awready(awready),
        .awaddr(awaddr),
        .wvalid(wvalid),
        .wready(wready),
        .wdata_out(wdata_out),
        .wstrb(wstrb),
        .bready(bready),
        .bvalid(bvalid),
        .bresp(bresp),
        .arvalid(arvalid),
        .arready(arready),
        .araddr(araddr),
        .rvalid(rvalid),
        .rready(rready),
        .rdata_in(rdata_in),
        .rresp(rresp)
    );
    axi_lite_slave slave (
        .clk(clk),
        .rst_n(rst_n),
        .awvalid(awvalid),
        .awready(awready),
```

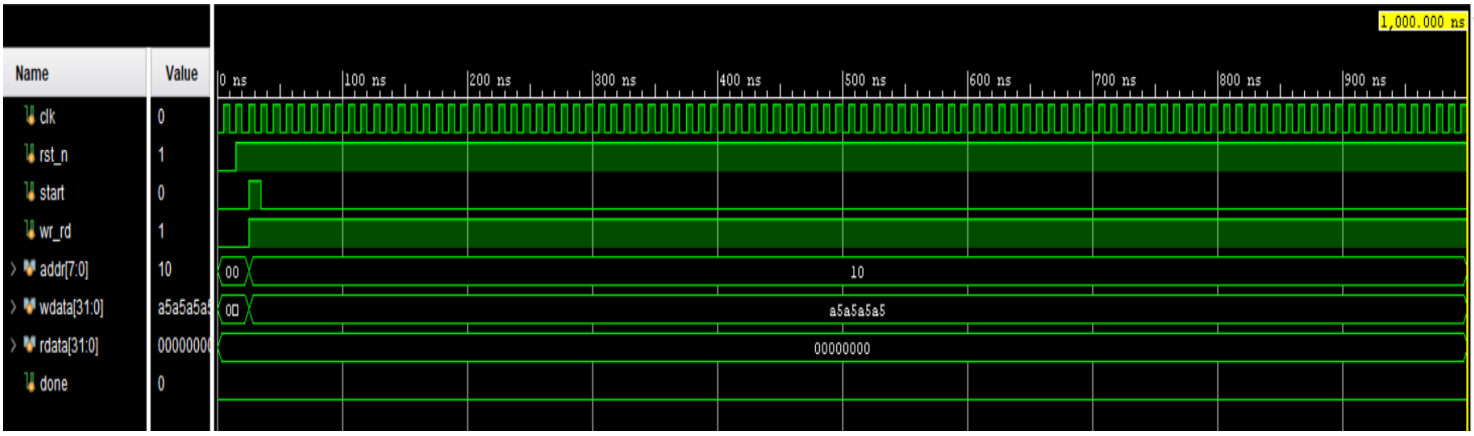
```
        .awaddr(awaddr),
        .wvalid(wvalid),
        .wready(wready),
        .wdata(wdata_out),
        .wstrb(wstrb),
        .bready(bready),
        .bvalid(bvalid),
        .bresp(bresp),
        .arvalid(arvalid),
        .arready(arready),
        .araddr(araddr),
        .rvalid(rvalid),
        .rready(rready),
        .rdata(rdata_in),
        .rresp(rresp)
    );
endmodule
```

- **TEST BENCH CODE:**

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 02
// TEST-BENCH FOR AXI LITE
module tb_top_axi_lite;
    logic       clk;
    logic       rst_n;
    logic       start;
    logic       wr_rd;
    logic [7:0]  addr;
    logic [31:0] wdata;
    logic [31:0] rdata;
    logic       done;
    top_axi_lite dut (
        .clk(clk),
        .rst_n(rst_n),
        .start(start),
        .wr_rd(wr_rd),
        .addr(addr),
        .wdata(wdata),
        .rdata(rdata),
        .done(done)
    );
    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end
    initial begin
        $dumpfile("axi_lite.vcd");
        $dumpvars(0, tb_top_axi_lite);
```

```
    end
    initial begin
        rst_n = 1'b0;
        start = 1'b0;
        wr_rd = 1'b0;
        addr = 8'd0;
        wdata = 32'd0;
        #15 rst_n = 1'b1;
        #10;
        wr_rd = 1'b1;
        addr = 8'h10;
        wdata = 32'hA5A5A5A5;
        start = 1'b1;
        #10 start = 1'b0;
        wait(done);
        #10;
        wr_rd = 1'b1;
        addr = 8'h20;
        wdata = 32'h12345678;
        start = 1'b1;
        #10 start = 1'b0;
        wait(done);
        #10;
        wr_rd = 1'b0;
        addr = 8'h10;
        start = 1'b1;
        #10 start = 1'b0;
        wait(done);
        #10;
        wr_rd = 1'b0;
        addr = 8'h20;
        start = 1'b1;
        #10 start = 1'b0;
        wait(done);
        #20;
        $finish;
    end
endmodule
```

❖ WAVE FORM OUTPUT:



❖ SCHEMATIC OUTPUT: